Setup and Configuration of a LAN using Switch & Ethernet

Exp no:4 Date:

Aim:

To set up and configure a LAN using switch and Ethernet cables.

Theory:

Definition: A LAN connects computers and devices within a limited physical area to enable fast, secure data transfer and resource sharing. A LAN switches acts as the central connecting device.

IP Address Configuration Table:

| Device | Assigned IP Address | Subnet Mask | Interface Connection |
|---|---|---|---|
| PC1 | 10.1.1.1 | 255.0.0.0 | Ethernet Port to Switch Port |
| PC2 | 10.1.1.2 | 255.00.0 | Ethernet Port to Switch Port |
| PC3 | 10.1.1.3 | 255.00.0 | Ethernet Port to Switch Port |
| PC4 | 10.1.1.4 | 255.0.0.0 | Ethernet Port to Switch Port |
| Switch | 10.1.1.5 | 255.0.0.0 | Management web Interface. |

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Step-by-Step Procedure:

① Topology Design & Hardware Connection:

* Plan the topology layout based on lab equipment locations

* Connect each of the 4 computers to the network switch using standard Ethernet cables

② Assigning IP Addresses to Computers:

* Log in as Administrator or Owner on each client computers.

* Open Network and Internet Connections

* Right Click LAN connection → Select Properties → Select TCP/IPv4 → click properties

* Select "Use the following IP Address" and enter the static IP and subnet mask according to the table above.

③ Configuring the Network Switch:

* Connect a PC to the switch via Ethernet cable and open a Web browser

* Enter the switch's default IP address in the address bar to open its web interface login page

④ Testing Connectivity:

* Open Command prompt on any PC and run the ping command to verify physical and logical communication between devices and the switches

⑤ & Resource Sharing Verification:

* Select a folder on one computer → Right-click and choose properties → Go to the sharing tab → Share it with everyone on the LAN.

* Access the shared folder from other computers across the network to confirm successful setup.

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